MOBILE DEVELOPER → ON-DEVICE AI

ARSENIUS PURBANDONO

I build Flutter apps that field officers use every day, then teach their cameras to read license plates. Offline, on cheap phones.

Bandung, Indonesia · Regional Revenue Agency of West Java · since 2023

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NUMBERS, NOT ADJECTIVES

3+ years shipping Flutter apps	**56→76%** OCR exact match, ML Kit → my model	**−30MB** APK size after removing ML Kit
4× faster image crop, moved off the UI thread	**1.1M** parameters, 2.3 MB TFLite	**494** locked test plates, never seen in training

HOW A PHONE READS A PLATE

The ANPR pipeline inside Panah Pasopati. Every step runs on

the device, no server and no network.

01
CAMERA
Live frame from a low or mid-range Android phone

02
DETECT
YOLOv11n finds the plate and corrects rotation

03
CROP
Cropped once in an isolate, about 4× faster

04
COLOR
MobileNetV2 classifies plate color (4 classes)

05
READ
Custom CNN+CTC OCR, 76% exact match

06
LOOKUP
Plate number + color → vehicle tax record

HOME SCREEN

Panah Pasopati

In production · Flutter · TFLite · YOLO · Shorebird · Firebase · Kotlin

Vehicle and tax tracing for field officers. Point the camera at a plate and get the vehicle's tax record.

- Built the on-device ANPR module (Teman Pamor): YOLO detection, color classifier, custom OCR
- Patched the native Android camera plugin in Kotlin so the camera stops restarting 3–4×
- Hardened OTA updates: HTTPS only, newer versions only, SHA-256 support

ATOS PAMOR

In production · Flutter · MVC · Dio · Hive · flutter_map · Geolocator · OTA

Deep vehicle tracing for the special task force: data verification, history and administrative workflows for tax recovery.

- Photo evidence with watermark: officer ID, GPS and reverse-geocoded address
- One error-handling shape across all controllers, real backend messages shown to users
- Migrated the app to a new Flutter and Dart major version

SAKTI JAWARA

In production · Flutter · Device SDK · Thermal printer · Excel/PDF

Tax payment and data-change tool for officers at every Samsat service point in West Java.

- Integrated custom device SDKs: camera, thermal printer, second screen
- Excel and PDF export
- Keyboard shortcuts for fast counter work

Sakti Mobile

In production · Flutter · Camera · Excel/PDF

Mobile companion for verifier officers: tax payment and data-change verification.

- Camera integration for evidence
- Local data management
- Excel and PDF export

Samsat Digital

In production · Flutter · Kiosk · Device SDK

Self-service tax payment kiosk for the public, placed at Samsat branches across West Java.

- Custom SDK integration: camera, printer and other kiosk hardware
- Local data management for kiosk sessions

SEMAR

In production · Flutter · Android + Web · Multi-package · Hive

Water-tax recording and verification: officers record usage in the field, verifiers approve it.

- Modular packages: recording, verification, history
- Runs on Android and the web from one codebase
- Offline-first storage with Hive

Pajak Alat Berat

In production · Flutter · MVC

Heavy-equipment tax app for recording and tracing excavators, cranes and other heavy machinery.

- Feature development and maintenance in the shared Flutter architecture

Aing Linmas

Prototype · Flutter · Android + iOS · QR · PDF

Mobile version of a registration system for Linmas (community protection) members, from province down to village.

- 3-step registration with NIK check and auto-saved drafts
- Digital member card with QR, downloadable as image or PDF
- Mock data for now, backend not connected yet

MODEL LAB

Where the models get built: data, labeling, training, and evaluation that does not lie.

WHY I REPLACED GOOGLE ML KIT



ML Kit is a general-purpose OCR. On Indonesian plates it kept confusing letters and digits, like O and 0, so officers had to retype.

I locked a test set of 494 real plates first, split by plate so nothing leaks into training. Then I generated 30k synthetic plates in plate fonts, trained a small CNN+CTC reader and exported it to TFLite with a parity check.

It shipped only after it beat ML Kit on that locked set: 76% vs 56% exact match. Removing ML Kit also cut the APK by about 30 MB.

Plate detector

IN PRODUCTION

YOLOv11n → TFLite

mAP50 81.6%

Finds the plate in the frame, with rotation auto-correction for skewed field photos.

Plate OCR

IN PRODUCTION

CNN + CTC · 1.1M params · 2.3 MB

76% exact

Reads the characters. Beat Google ML Kit (56%) on 494 locked test plates.

Plate color

IN PRODUCTION

MobileNetV2 α0.35 · 96×96

~97% val acc

White, black, yellow, red. Red is still the weak class: few real samples, partly synthetic.

Anti-spoofing

BACKLOG

Binary CNN

–

Planned: reject photos taken of a screen (moiré, reflections) so evidence stays genuine.

SANDBOX

Practice projects outside work, to learn NLP and transfer learning.

Amazon Review Sentiment

87.25% acc · F1 0.87

TF-IDF · Logistic Regression · NLTK · Flask

Trained on 1.6M reviews, tested on 400k.

Casting Defect Inspection

91.8% val acc

MobileNetV2 transfer learning · Keras · Tkinter

Defective vs OK pump impellers, 1,016 industrial images.

LOG

EXPERIENCE

a76c0de Mobile Application Developer

HEAD

Regional Revenue Agency of West Java (Bapenda Jabar) · Feb 2023 – now

Flutter apps for tax and vehicle tracing, Samsat service points and field verification. Multi-package codebases (4–19 packages), MVC and Clean Architecture, Shorebird code push and hardened OTA updates. Built and shipped the on-device ANPR models.

5a1e922 Mobile Developer Intern

PT Safepedia Global Teknologi, Jakarta · Aug – Dec 2022

Native Android in Kotlin with Clean Architecture for PT Antam's mining operators app. Bug fixes and new features.

EDUCATION

B.Ed. Information Systems & Technology Education, Universitas Pendidikan Indonesia (2019–2023)

CERTIFICATIONS

- **BNSP Senior Mobile Programmer** · 2025-11
- **BNSP Junior Mobile Programmer** · 2025-08
- **EF SET English B2** (54/100) · 2024-06
- MSIB Mobile & Cloud Dev, Nongsa Digital Park · 2022
- MSIB Game Development, Agate · 2021–22

LANGUAGES

- Japanese: beginner, studying with Minna no Nihongo, aiming for JLPT N4
- English: B2 (EF SET 54/100)
- Indonesian: native

Looking for a software or AI engineering role in Japan. Available from the second half of 2027, needs work-visa sponsorship.

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